A03 Neural Network Zoo

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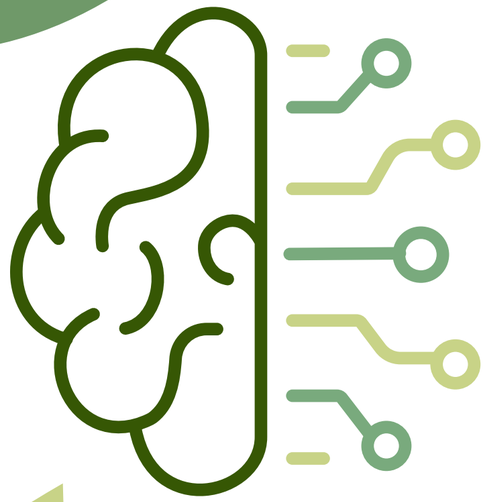
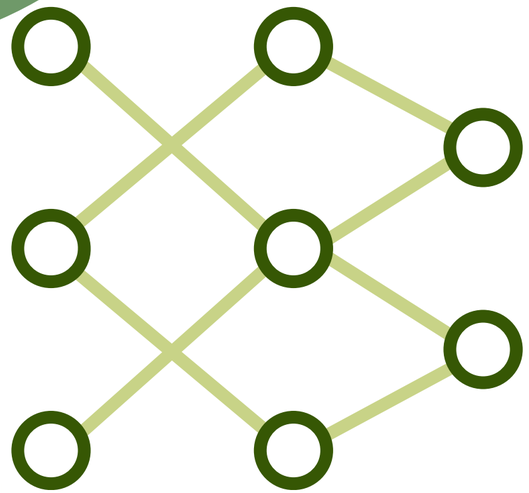
Introduction



Neural networks are based on how the neurons in our brain work. This algorithm can recognize patterns, learns from them, and make predictions by using simple neurons and layers. The prediction process uses weights and biases. These two control the influence and the values of each input that help the network make a decision. A neural network consists of different layers, and each one has different functions. The input layer is the one that receives raw data, the hidden layers consist of artificial neurons, and the output layer, where we can see the prediction.

Deep learning uses neural networks and multiple layers, so this subset of machine learning is more complex. Deep learning is similar to how our brain works; that's the reason it can solve complex problems such as image recognition and speech processing.

Structure of a Neuron



A neuron consists of:

- Inputs: numeric values that represent data.
- Weights: determine the importance of the input.
- Bias: helps adjust the output.
- Activation function: decides if the neuron should fire.

A neural network structure consists of:

- Input layer: receives the raw data.
- Hidden layers: Process the data, learn, and make a decision.
- Output Layer: Produces the final decision.

Neural Network Types:

- Convolutional Neural Network (CNN).
- Recurrent Neural Network (RNN).
- Long Short-Term Memory (LSTM).
- Generative Adversarial Network (GAN).

Welcome to the Neural Network Zoo!

In this presentation, the different neural networks are represented as animals. I highlighted the network structure, the way it works, applications, strengths, and limitations. Each animal represents a unique neural network.

I represented a Convolutional Neural Network (CNN) with a cheetah, a Recurrent Neural Network (RNN) with a raccoon, a Long Short-Term Memory network (LSTM) with a lemur, and a Generative Adversarial Network (GAN) with a fox.

CNN CHEETAH

Convolutional Neural Network

Network Structure

- Convolutional layers
- Pooling layers
- Fully connected layers
- Output layer

How It Works

The CNN scans images using filters that detect edges, shapes, and textures. As data moves deeper into the network, it learns more complex visual features to make accurate predictions.

Application

- Image classification
- Facial recognition
- Medical imaging
- Object detection

Strengths

- Excellent at image and video tasks
- Automatically learns spatial features
- Efficient pattern detection

Limitations

- Requires large labeled datasets
- Computationally expensive to train

Why a Cheetah to represent CNNs?

Cheetahs are fast and have excellent vision. CNNs are optimized for fast and accurate visual pattern recognition.

RNN RACCOON

Recurrent Neural Network

Network Structure

- Input layer
- Recurrent hidden layer(s)
- Output layer
- Feedback loops

How It Works

An RNN processes data one step at a time. It passes information from previous steps forward, allowing the network to remember past inputs when making predictions.

Applications

- Speech recognition
- Language translation
- Text generation
- Time series prediction

Strengths

- Handles sequential and time-based data
- Maintains short-term memory
- Simple and flexible architecture

Limitations

- Struggles with long-term dependencies
- Can suffer from vanishing gradients

Why a Raccoon to represent RNNs?

Raccoons are known for their memory and problem-solving skills. RNNs remember previous inputs, making them ideal for sequential data.

LSTM LEMUR

Long Short - Term Memory Network

Network Structure

- Input layer
- LSTM cells
- Memory cell state
- Output layer

How It Works

LSTM networks use special gates to decide what information to remember, forget, or update. This allows the network to retain important information over long sequences.

Applications

- Speech recognition
- Language translation
- Text prediction
- Time series forecasting

Strengths

- Handles long-term dependencies
- Reduces vanishing gradient problems
- More stable than basic RNNs

Limitations

- More complex than standard RNNs
- Slower training due to gate mechanisms

Why a Lemur to represent LSTMs?

Lemurs are alert and have strong memory skills. LSTMs are designed to remember important information for long periods of time.

GAN FOX

Generative Adversarial Network

Network Structure

- Generator network
- Discriminator network
- Adversarial training loop
- Output layer

How It Works

GANs use two networks that compete with each other. The generator creates fake data, while the discriminator tries to detect whether the data is real or fake. Over time, both networks improve.

Applications

- Image generation
- Deepfake creation
- Data augmentation
- Art and creative design

Strengths

- Generates realistic synthetic data
- Encourages creative outputs
- Powerful unsupervised learning method

Limitations

- Difficult to train and stabilize
- Can produce unrealistic results if poorly tuned

Why a Fox to represent GANs?

Foxes are clever and strategic. GANs generate realistic data by learning to trick the discriminator, improving through competition.

References

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Thank You

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